

Jack Rademacher

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SUMMARY

Experienced engineer, researcher, and team leader specializing in wireless communication systems, digital signal processing, embedded hardware and firmware, power electronics, RFIC design, and underwater acoustics. During my PhD, I led multi-person teams to design and build novel underwater systems for sensing, communication, and networking. I combine deep technical expertise with communication and presentation skills developed over years interaction with academia and industry.

RESEARCH EXPERIENCE

Signal Kinetics at the MIT Media Lab — Massachusetts Institute of Technology

Graduate Research Assistant, advised by Fadel Adib

September 2021 — May 2026

- Actively managing multiple research projects on persistent oceanographic sensing and underwater localization
- Deployed an integrated, wireless conductivity, temperature and depth sensor to Woods Hole Oceanographic Institute. Live data available at <https://sk-exp-server.mit.edu>
- Led a four-person team to design, build, and test a first-of-its-kind passive underwater retrodirective acoustic beamformer
- Managed multiple undergraduates and masters of engineering students to design a sub- μ W underwater temperature sensor
- Designed, built, and tested an ultra-low power platform for battery-free wireless imaging of underwater environments
- Designed hardware platforms and signal processing algorithms for ultra low-power acoustic backscatter communication

WORK EXPERIENCE

MITRE — Bedford, MA

May 2023 — August 2023

Sensor Systems Engineering Intern, L156 (Underwater Systems) / L152 (Radio Signals Intelligence)

- Developed a mathematical model of the underwater backscatter link budget combined with the multipath acoustic channel
- Developed and wrote software for a network-layer packetization standard for underwater acoustic networks
- Analyzed and simulated array processing & beam steering techniques such as MVDR, MPDR, MUSIC, CBF, and power inversion for HF (3-30 MHz) radio antenna arrays

Analog Devices — Colorado Springs, CO

May 2021 — August 2021

Microwave Design Engineering Intern, Aerospace & Defense (ADEF) Business Unit

- Designed an integrated 6-14 GHz, 128 ps, 6 bit true-time delay unit for beamforming ASICs on a 45nm RF-SOI process
- Designed and simulated a time delay architecture not previously implemented by my team that improved layout area and time delay density by up to 45%
- Designed and simulated bypass networks for mimicking S_{21} of delay units and providing a flat differential time delay across 2ps, 4ps, 8ps, 16ps, 32ps, and 64ps units

Analog Devices — Colorado Springs, CO

May 2020 — August 2020

Microwave Product Engineering Intern, Aerospace & Defense (ADEF) Business Unit

- Created a test setup for characterizing IP₃ of beamforming ICs from 2-33GHz through PCBs or an RF probe
- Collected data on worst/average case input and output IP₃ to within ± 0.35 dBm of actual value across frequency & power to compare with designers' simulations
- Communicated and displayed summer work to upper level management, showing how my test bench is accurate, easy to use, and can be used over wide range of bandwidths, power levels, and device form factors

Apple — Sunnyvale, CA

January 2019 — August 2019

Electrical Engineering Intern, Special Projects Group (SPG)

- Designed the schematic and layout for a "wireless oscilloscope" PCB to measure and log voltage, current and temperature
- Wrote C to live-stream ADC samples to an SD card and wirelessly to a python application using the MQTT protocol
- Created a concurrent embedded application using FreeRTOS to manage ADC sampling, WIFI connectivity, wireless streaming & device control, SD card reading/writing, and RTC initialization using network time

EDUCATION

Massachusetts Institute of Technology — Cambridge, MA

September 2021 — May 2026

M.S./Ph.D. Electrical Engineering and Computer Science

University of Michigan — Ann Arbor, MI

September 2017 — May 2021

B.S.E. Electrical Engineering, Summa Cum Laude

SKILLS

Programming Languages: C/C++, Python, MATLAB, Java, HTML/CSS, JavaScript (React, ReactNative, NodeJS)

Software Packages: Altium Designer, Keysight ADS, Solidworks, LTSpice, Cadence Allegro, Cadence Virtuoso, NI Labwindows/CVI, NI Max, COMSOL